

Chronos branch management

- Each branch owns a shrinking interval in a 1-D integer space
- row versions get intervals from the branch that wrote them
- updates split old row intervals to preserve isolation

Branch management

Allocation Po



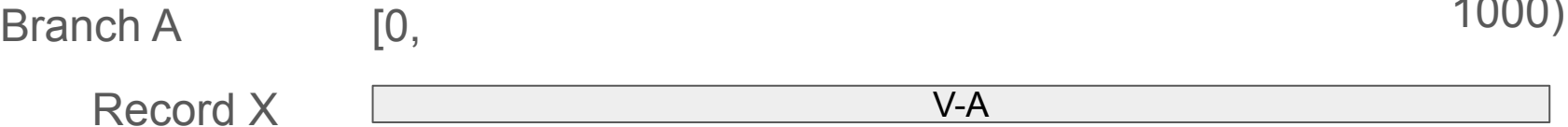
Branch A

[0,

1000)

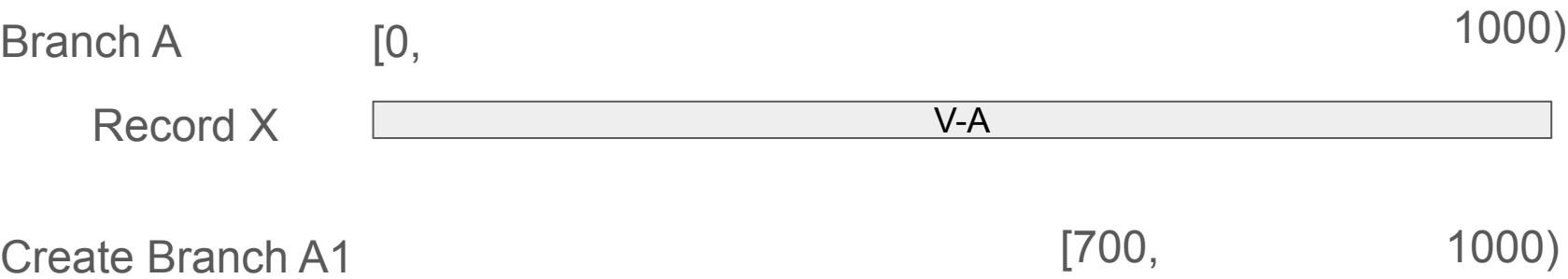
Branch management

Allocation Po
↓



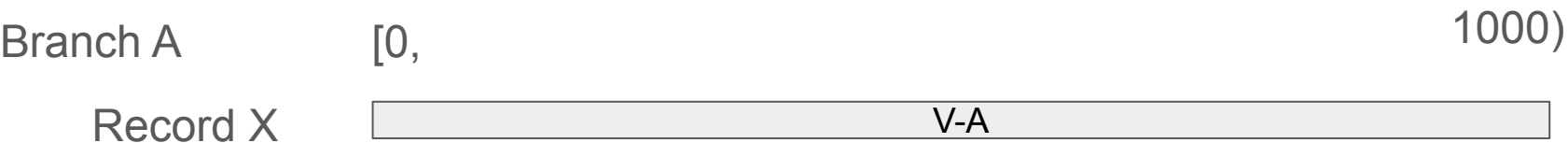
Branch management

Allocation Pointer
↓



Branch management

Allocation Pointer
↓

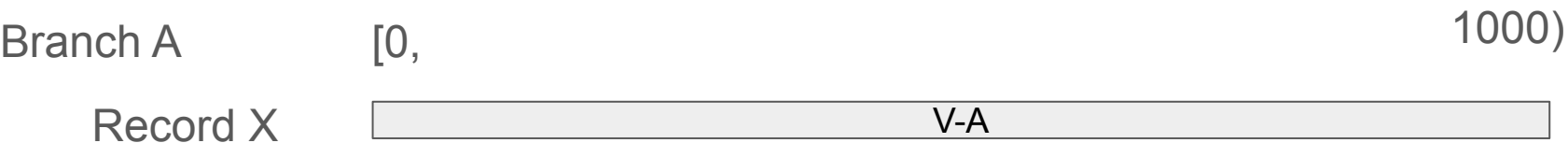


Branch A1 [700, 1000)

Create Branch A2 [400, 700)

Branch management

Allocation Pointer
↓



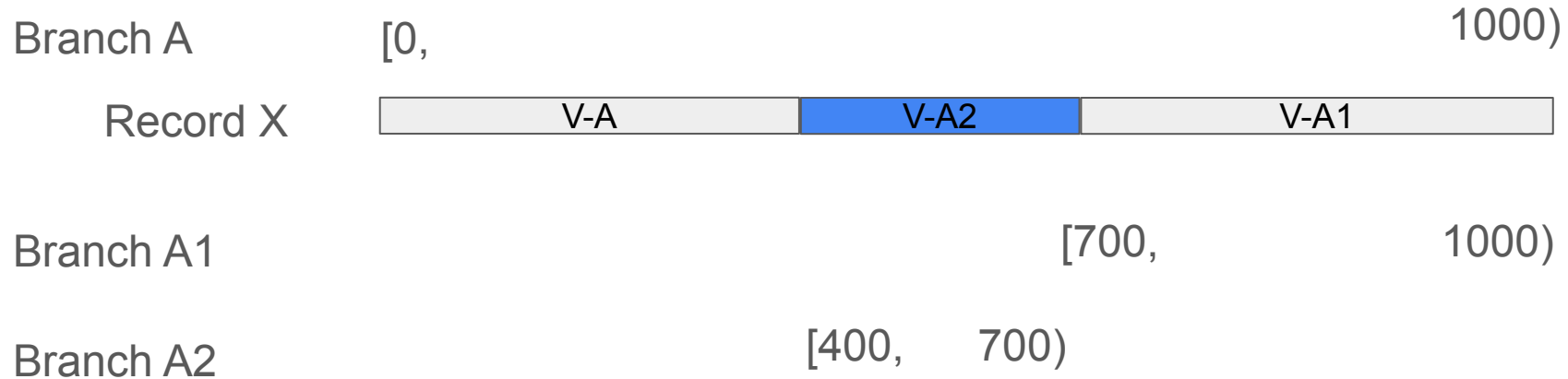
Branch A1 [700, 1000)

Create Branch A2 [400, 700)

Branch A2 Update X

Branch management

Allocation Pointer

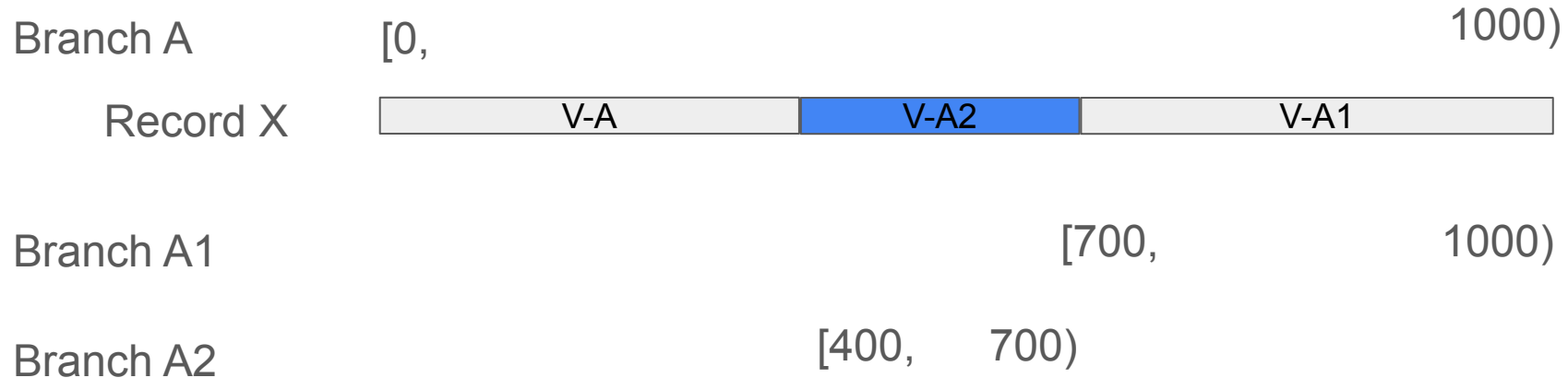


Branch A2 Update X,

Duplicate X into 3 non-overlapping versions
before A2 can physically update X

Visibility Predicate

Allocation Pointer



Visibility Predicate: **row.lo** <= **b.read_point** < **row.hi** **AND** **row.deleted** = false

Branch A.read_point: any point in [0, A1.allocation_pointer)

Branch A1.read_point: any point in [700, A1.allocation_pointer)

Branch A2.read_point: any point in [400, A2.allocation_pointer)